TACROLIMUS

Formulations NOT interchangeable

Nephrotoxic; new-onset diabetes

CHEMICAL NAME Macrolide lactone immunosuppressant (FK506) from *Streptomyces tsukubaensis* ($C_{44}H_{69}NO_{12}$)

CLASS Calcineurin inhibitor immunosuppressant

TRADE NAMES Prograf · Advagraf (once daily) · Envarsus · Protopic (ointment)

THERAPEUTIC USES / INDICATIONS

- Prevention and treatment of transplant rejection (kidney, liver, heart) — first-line calcineurin inhibitor.
- Atopic dermatitis (topical ointment).
- Lupus nephritis, myasthenia gravis (off-label).

ADMINISTRATION

- **PO** on an empty stomach (1 h before or 2–3 h after food), same times daily.
- Prograf twice daily; Advagraf/Envarsus once daily in the morning — never switch without specialist (dosing errors → rejection/toxicity).
- **IV** continuous infusion (rare).
- Topical ointment twice daily (no occlusion).

DOSAGES

- **Kidney transplant:** 0.1–0.3 mg/kg/day orally in 2 doses, adjusted to trough levels (e.g. 5–15 ng/mL per time post-transplant).
- **Liver transplant:** 0.1–0.2 mg/kg/day.
- **Ointment:** 0.03% (children) or 0.1% twice daily.

PHARMACOKINETICS

- **Peak:** 1–3 h. Variable bioavailability (~25%); food ↓ absorption.
- Metabolised by CYP3A4/5 (CYP3A5 expressers need higher doses).
- Excreted in bile. **t½:** 12–16 h.

NURSING MANAGEMENT

- Trough levels 12 h (or 24 h) post-dose before next dose.
- Monitor creatinine, K^+, Mg^{2+}, glucose, BP, tremor.
- Check brand and formulation at every transition of care.
- Infection surveillance; sun protection.

PHARMACODYNAMICS

- Binds **FKBP-12** → inhibits **calcineurin** → ↓ IL-2 and T-cell activation (~10–100× more potent than ciclosporin).

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Tremor, headache, insomnia.
- Diarrhoea, nausea.
- Hypertension; hair loss (not hirsutism).

ADVERSE EFFECTS (SERIOUS)

- Nephrotoxicity.
- New-onset diabetes after transplant.
- Neurotoxicity (seizures, PRES); hyperkalaemia, hypomagnesaemia.
- Infections (CMV, BK virus); malignancy (skin, PTLD); QT prolongation.

CONTRAINDICATIONS & CAUTIONS

- Hypersensitivity to tacrolimus or macrolides.
- Live vaccines.
- **Caution:** QT prolongation, pregnancy, hepatic impairment.

KEY INTERACTIONS

- **↑ levels:** clarithromycin, azoles, diltiazem, verapamil, grapefruit, protease inhibitors.
- **↓ levels:** rifampicin, carbamazepine, phenytoin, St John's wort.
- **Nephrotoxic drugs, K^+-sparing diuretics:** additive toxicity.

PATIENT EDUCATION

- Take at exactly the same times on an empty stomach.
- Never change brand or formulation yourself.
- Avoid grapefruit, St John's wort; check new medicines.
- Report tremor, high glucose symptoms, fever, reduced urine.

KEY POINTS

- Calcineurin inhibitor; first-line in transplants.
- Nephrotoxicity, tremor, diabetes.
- Trough monitoring; formulations not interchangeable.
- CYP3A4 interactions.

SOURCES CHECKED **[856]** Adoport 0.5 mg Hard Capsules (UK SmPC) **[857]** Prograf 5mg/ml Concentrate for Solution for Infusion (UK SmPC) **[858]** Prograf, oral, IV (US label) **[859]** Tacrolimus, IV (US label)

Full references p. 432–457